

EGM 5421: Structural Dynamics - Project

Instructions:

- (1) The Project is open book and open notes. You may work on these problems at your home or any convenient place. The project is a group activity with no more than 2 partners.
- (2) You may not discuss this test with anyone (other than your project partner). If you are unclear about a problem, you may contact me via email. **I will answer questions received in email by 5pm Friday, April 18, 2025.**
- (3) The test should be turned in by 11:50 p.m. on Friday, April 25, 2025.

Name:.....

1. A cellular telecommunication tower is to be analyzed for a variety of dynamic loads.
 - A. First to identify its dynamic properties modeled as a SDOF, a lateral force of 10000 lb is applied at the top of the tower resulting in a horizontal displacement of 1 inch. A cable attached to the top of the tower is suddenly severed resulting in a free vibration. At the end of 20 complete vibration cycles, the time is 200 sec and the amplitude is 0.02 inch. Find effective stiffness, effective mass, undamped natural frequency and damping ratio for the SDOF system.
 - B. The tower must survive an extreme natural event, such that the imposed loading can be modeled as a superposition of a harmonic, triangular and square wave loading whose are characteristics are given as follows:
 - 1) harmonic loading – amplitude= 10kips, period=0.1s and duration=5s,
 - 2) triangular loading – amplitude= 15kips, period=0.12s and duration=4s,
 - 3) square loading – amplitude= 5kips, period=0.15s and duration=5s,Using Nonlin, find the imposed wave functions and its FFT. Find the response of the system assuming that the system is linear. Is the tower designed sufficiently well if the design criterion is a maximum deflection of 0.5 inch.
 - C. If the tower is not sufficiently well designed, then suggest alternatives to the owner.
 - D. What will be the response of the original system if the tower has a yield point is 0.5kips and the secondary stiffness is 1 kip/in. What is the permanent deflection of the tower. Will the tower survive this event.
 - E. The tower is also subjected to an impact loading of 1000 kips of duration 0.05s. What will be the response, first assuming linear behavior and, subsequently, assuming nonlinear behavior.